

Isomerism — Class Notes from Pankaj Sir (PW Lakshya JEE 2027)

☀ 3 Golden Rules for Organic Chemistry (OC)

Before starting Isomerism, Sir gave three strict rules for success in OC [1]: 1. **Write Your Own Notes (No Printed PDFs):** Organic Chemistry is learned through the tip of your pen [1]. 2. **No Backlogs, No Excuses:** Today's lecture must be revised today. A backlog in OC is an invitation to failure [1]. Daily routine: 30 Min Rev + 1 hr solving (DPPs, P.S, KBCC) [1]. 3. **Action Over Theory:** Theory is only 30% of the battle. The remaining 70% is won by solving DPPs, modules, and PYQs [1]. * **Books Recommended:** Paula Bruice, Clayden (for Advanced), and Peter Sykes [2].

🧩 Degree of Unsaturation (D.U.)

Also known as **I.H.D.** (Index of Hydrogen deficiency), **U.I.** (Unsaturation Index), or **D.B.E.** (Double bond equivalent) [3]. * **Definition 01:** Number of H_2 molecules required to convert a molecule (multiple bond) into an open chain saturated compound [4]. * **Definition 02:** How many bonds are cleaved to form an open chain saturated compound [5].

💡 POPU BOX TRICK for D.U. Calculation:

- **1 π bond = 1 D.U.** [4].
- **1 Ring = 1 D.U.** [5].
- *Example:* Benzene has 1 ring and 3 π bonds $\rightarrow$ D.U. = 4. A molecule with a benzene ring and an ester group has D.U. = 5 [6]. A molecule with a cyanide ($-CN$), a double bond, and a ketone has D.U. = 4 [6].

Formula if Molecular Formula (M.F.) is given: $DU = (C + 1) - \frac{(H + X - N)}{2}$ [7] * (C = Carbon, H = Hydrogen, X = Halogen, N = Nitrogen) * *Example 1:*

$C_8H_8Cl_2 \rightarrow DU = (8 + 1) - \frac{(8 + 2 - 0)}{2} = 9 - 5 = 4$ [8]. * *Example 2:*

$C_8H_{10}O_2 \rightarrow DU = (8 + 1) - \frac{10}{2} = 9 - 5 = 4$ [8]. (Note: Oxygen is ignored in the formula).

🧭 Introduction to Isomerism

- **Definition:** Isomers have the same molecular formula but different physical and chemical properties [9].

- Compounds having the same molecular formula but **different connectivity** (structural isomerism) or **different 3-D relative arrangement** (stereoisomerism) are known as isomers [9].
- Important Conditions:** Isomers are those which can be separated at room temperature [9]. Isomers **always** have the same Degree of Unsaturation (DU) [9].

Classification Tree of Isomerism [10]:

- Structural Isomerism:** Chain, Position, Functional, Metamerism, Tautomerism, Ring-chain [10].
- Stereoisomerism:** * Configurational (Geometrical, Optical) [10]. * Conformational [10].



1. Structural Isomerism

Same molecular formula, different connectivity of atoms [11]. (e.g., Butan-1-ol vs Butan-2-ol, where the -OH group changes position) [11].

A. Chain Isomerism (C.I.)

- Definition:** Compounds which have the same molecular formula but have different parent or side chain [12].
- Example:** Pentane (C_5H_{12}) vs 2-methylbutane vs 2,2-dimethylpropane. All have the same M.F. but different parent chain lengths [12].

B. Position Isomerism (P.I.)

- Definition:** Compounds which have the same molecular formula but have different position of branch or F.G. (double or triple bond) [13].
- Example:** 1-Chloropropane and 2-Chloropropane [13]. 1-Propanol and 2-Propanol [13].

C. Metamerism

- Definition:** Compounds which have the same molecular formula but have different arrangement of atoms or groups from either side of a bridging functional group [14].
- Bridging Functional Groups:** -O- (Ether), -S- (Thioether), -C(=O)- (Ketone), -NH- (2° Amine), -S(=O)₂-O- (Sulfonate), -C(=O)-O- (Ester) [14].
- Example:** Diethyl ether ($C_2H_5 - O - C_2H_5$) and Methyl propyl ether ($n - C_3H_7 - O - CH_3$) [15].

D. Functional Isomerism (F.I.)

- Definition:** Compounds which have the same molecular formula but have different functional groups [16].

- **Important Functional Isomer Pairs [17]:**
- Alcohols and Ethers [17] (e.g., $\text{CH}_3\text{CH}_2\text{OH}$ and $\text{CH}_3 - \text{O} - \text{CH}_3$) [16].
- Aldehydes and Ketones [17].
- Cyanides and Isocyanides [17].
- Primary (1°), Secondary (2°), and Tertiary (3°) Amines [17].
- Acids and Esters [17].
- Diene and Alkyne (yne) [17].

E. Ring-Chain Isomerism

- **Definition:** Compounds which have the same molecular formula, but one compound is an open chain & the other is cyclic [18].
- **Example:** Propene ($\text{CH}_3 - \text{CH} = \text{CH}_2$) and Cyclopropane (triangle ring), both having M.F. C_3H_6 [18].

F. Homologous Series

- **Definition:** A group or series of organic compounds each containing a characteristic functional group forms a homologous series, and members are called homologues [19].
- **Characteristics:** Same Functional group, Different Molecular formula, Different Molecular mass, Differ by a $-\text{CH}_2-$ group [19].
- **Example:** Alkanes ($\text{CH}_4 \rightarrow \text{C}_2\text{H}_6 \rightarrow \text{C}_3\text{H}_8$) [20].

2. Conformational Isomerism

- **Definition:** Arises due to free rotation around C-C sigma (σ) bond [21].
- There are infinite conformers, and the energy for rotation (Rotational Energy) around the sigma bond is available at room temperature [21].
- **Total conformer of methane is zero** [21]. Also known as Rotamers [21].
- 💡 **POPU BOX:** Stereoisomers separated by a relatively low energy barrier (<60 kJ/mol) are easily interconvertible at ambient temperature. **Conformers are not true isomers** [21, 22].

Types of Strains (Destabilization) [23]:

1. **Torsional Strain:** Arises due to repulsion between bond pair electrons (eclipsing bonds) [23].
2. **Vander Waal Strain:** Arises due to repulsion of front & back atoms/groups due to steric factor (bulkiness) [23].

I. Conformations of Ethane ($CH_3 - CH_3$)

- **Forms:** Eclipsed ($\theta = 0^\circ$) $\rightleftharpoons$ Staggered ($\theta = 60^\circ$) [24]. All intermediate forms are called **skew forms** [24].
- **Stability:** Staggered > Skew > Eclipsed [24].
- **Energy Barrier:** ~ 3 kcal/mol (12 kJ/mol) between Eclipsed and Staggered [25].

II. Conformations of Butane

- **Forms & Dihedral Angles (θ) [26-28]:**
- Fully Eclipsed (FE): $\theta = 0^\circ$ (Max energy, least stable) [26, 28].
- Gauche (G): $\theta = 60^\circ$ [26, 28].
- Partially Eclipsed (PE): $\theta = 120^\circ$ [26, 28].
- Anti (A): $\theta = 180^\circ$ (Min energy, most stable) [26, 28].
- **General Stability Order:** Anti > Gauche > Partially Eclipsed > Fully Eclipsed [26].

 **EXCEPTION: The Gauche Effect [29, 30]** If hydrogen bonding is present in Gauche conformations, it becomes **more stable than anti conformations** [29]. * Groups that show this effect: OH, NH₂, F, NO₂ [29]. * *Example Compound:* Ethylene glycol (Ethane-1,2-diol) or 2-nitroethanol [31-33]. * **Stability Order (with Gauche effect):** Gauche > Anti > Partially Eclipsed > Fully Eclipsed [29].

III. Conformations of Cyclohexane [34]

A cyclohexane ring is in a continuous state of flipping into different shapes: Chair, Twist boat, Boat, Half chair [34]. * **Stability Order:** Chair > Twist boat > Boat > Half chair [35]. * **Energy Barrier:** Transformation of chair form to other forms takes about 37.8 - 46.0 kJ/mol [35]. * **Ring Flip:** Chair conformations interconvert through bond rotation. During interconversion, equatorial hydrogens become axial and vice versa [36]. * **Flag-pole Interactions:** The Boat form is less stable than the chair form because it suffers from flag-pole interactions and torsional strain [37, 38].

 **Principle of Maximum Equatoriality (POME) [39]:** In the chair form, for maximum stability: 1. Maximum no. of groups must be present on equatorial positions [39]. 2. The **bulky group MUST be placed on the equatorial position** to avoid 1,3-diaxial interaction repulsions [39].

 **Exception to POME (1,3-diaxial interaction) [40]:** Normally, 1,3-diaxial positioning causes severe repulsion (e.g., between bulky methyl groups), making it less stable [40]. However, if **Hydrogen bonding** occurs (e.g., between two axial -OH groups), it causes *attraction*, making the 1,3-diaxial (cis) conformation more stable than the di-equatorial one [40].



3. Geometrical Isomerism (G.I.)

- **Definition:** Arises due to **Restricted rotation** around C=C, C=N, N=N, and Cycloalkanes etc. [41, 42].
- G.I. are **not interconvertible** at room temperature because of the high π -bond dissociation energy (~62 Kcal) [41].
- G.I. are also known as **diastereomers** (they are not mirror images). The areal distance must be different ($l_1 \neq l_2$) [41].

Condition for Exhibition of G.I.: The restricted site MUST have **different groups** attached to each atom [43]. (If one carbon of the double bond has two identical groups, it will NOT show G.I. [44]).

G.I. in Cycloalkanes [45, 46]: 1. At least two sp^3 carbons must be present in the ring [46]. 2. Minimum two groups must be present on those carbons [46]. (*Note: Cycloalkenes will only exhibit G.I. across the double bond if the ring size is ≥ 8 carbons [47].*)

Types of Geometrical Isomerism [48]:

1. Cis / Trans [49]

- **Cis:** Same groups are on the *same side* [49].
- **Trans:** Same groups are on the *opposite side* [49].

2. Syn / Anti System (Type 02) [50]

Used specifically for C=N and N=N systems. * **Aldoximes ($R - CH = N - OH$):** If the -OH and the -H are on the **same side**, it is **Syn**; otherwise, **Anti** [50]. * **Ketoximes ($R, R' - C = N - OH$):** Named based on the alkyl group. Example: *Syn-Ethyl methyl oxime* vs *Anti-Ethyl methyl oxime* [51]. * **Azo Compounds ($N = N$):** If the lone pairs (or attached groups) are on the same side, it is **Syn**, otherwise **Anti** [52].

3. E / Z System (C.I.P. Rules) [53]

Used when 4 different groups are attached to the restricted rotation site. * **Z (Zusammen):** Higher priority groups on the **same side** [53]. * **E (Entgegen):** Higher priority groups on the **opposite side** [53].

Cahn-Ingold-Prelog (C.I.P.) Priority Rules [53-58]: * **Rule 01:** Greater the atomic number, higher will be the priority [53]. * **Rule 02:** If the atomic number of the first atom is the same, then go to the next atom and apply rule 01 [55]. * **Rule 03:** In the case of isotopes, higher mass number will be the higher priority (e.g., Deuterium (D) > Hydrogen (H)) [56]. * **Rule 04 (Multiple Bonds):** Treat double/triple bonds as multiple single bonds to that atom. E.g., $-C \equiv C - H$ is treated as a Carbon bonded to 3 other Carbons. **Real > Hypothetical** [57, 58].



Physical Properties of Cis-Trans Isomers [59]

When comparing Cis vs Trans isomers, Sir gave these general trends [59-61]:

1. **Dipole Moment (μ):** Cis > Trans. Trans is often symmetric, bringing μ closer to or exactly 0 [60].
2. **Boiling Point (B.P.):** $B. P. \propto Dipole Moment$. Because Cis is more polar, **Cis > Trans** [60, 62].
3. **Crystal Lattice Energy (C.L.E.):** Amount of energy required to break 1 mol of crystal. Because Trans is more symmetrical, it packs better. **Trans > Cis** [60, 63].
4. **Melting Point (M.P.):** $M. P. \propto C. L. E.$. Trans isomers have higher melting points due to symmetrical packing. **Trans > Cis** [61, 63, 64].
5. **Stability:** Trans > Cis (due to less steric hindrance between bulky groups on opposite sides) [61, 65].
6. **Solubility:** Cis > Trans (polar dissolves in polar solvents) [61].

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